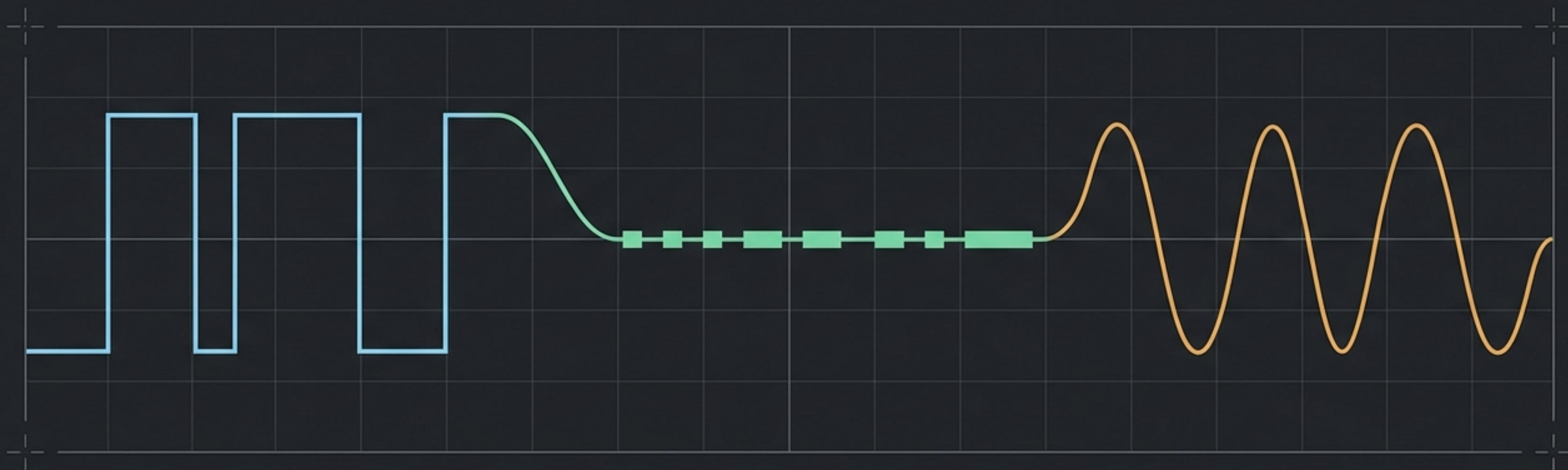




The Physics of Data Transit

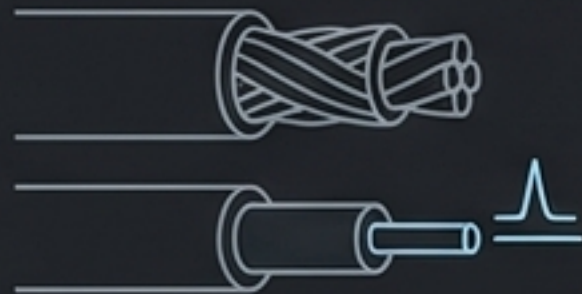
Signals, Latency, and the True Bottlenecks of Networking

The constraints of physics dictate network behavior.

Higher-level network engineering operates entirely within the boundaries of basic physics. Light speed, electrical interference, and spatial volume are the invisible forces shaping every byte transmitted.

1. Physical Form

Copper, Light, and Phase.



PHASE OFFSET: $+45^\circ$
 λ : 1558nm
 $C = 3 \times 10^8$ m/s
MATERIAL: SiO₂ CORE
IMPEDANCE: 1000

2. Environmental Reality

Noise and Attenuation.



ATTENUATION (dB/km)
DISTANCE

NOISE FLOOR: -110dBm
SNR: 12dB
INTERFERENCE: EM, RFI
POWER LOSS: -45% @ 5km

3. Anatomy of Delay

Time across space and hardware.



PROPAGATION DELAY: 5ms/1000km
SERIALIZATION: 0.2ms
QUEUEING DELAY: 1.5ms
PROCESSING: 50μs
TOTAL LATENCY: >6.75ms
GEOSPATIAL PATH: 12,000KM

4. Protocol Bottleneck

The Bandwidth-Delay Product.



$BDP = BANDWIDTH \times RTT$
BUFFER SIZE > BDP

TCP WINDOW: 64KB
LINK CAPACITY: 10Gbps
RTT: 10ms
OPTIMAL WINDOW: ~12.5MB
THROUGHPUT LIMIT: TCP FLOW CONTROL

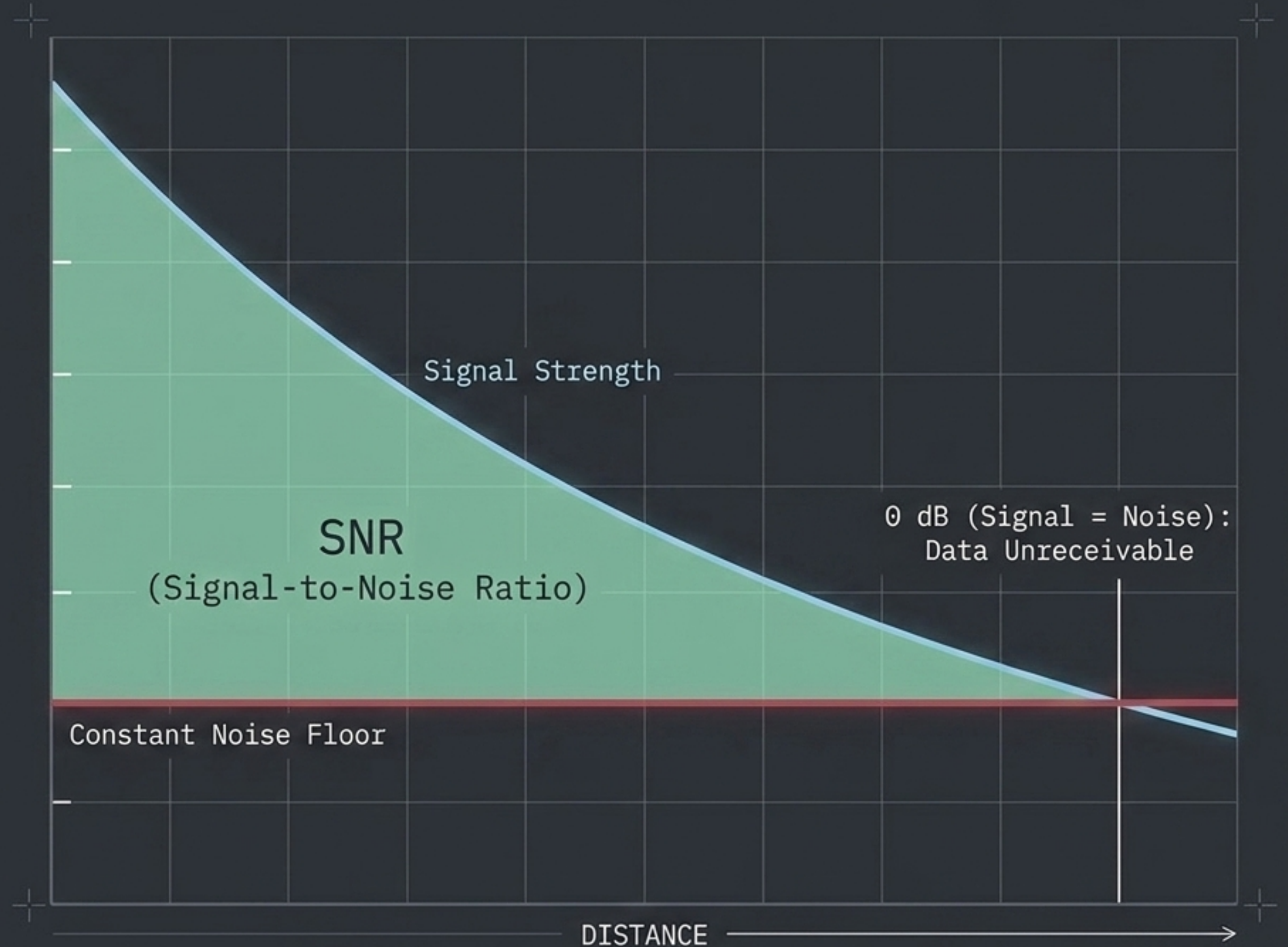
One Logical Bit, Three Physical Realities

The upper layers of networking never see physical signals. At Layer 1, the abstract concept of a bit requires completely different physical manifestations.



The Absolute Reality of Noise

Absolute signal strength is irrelevant. The gap between the signal and the background noise dictates whether a bit survives the environment.



Wi-Fi Modulation Fallback

Why does Wi-Fi slow down at the edge of coverage? Because as SNR drops, the access point must fall back to simpler, slower physical modulation to ensure the bits remain legible through the noise.

It changes the language, not just the volume.

	SNR (dB)	Modulation Scheme	Speed Impact
	≥ 25	64-QAM 5/6	Max Speed
	≥ 15	64-QAM 2/3	Moderate
	≥ 9	QPSK 3/4	Slow
	< 5	BPSK 1/2	Minimum Reliable

The Anatomy of Latency

'Ping' is not a single variable. It is the sum of four distinct physical and computational constraints.

$$\text{RTT} = (\text{Propagation} + \text{Transmission} + \text{Processing} + \text{Queuing}) \times 2$$

Propagation

Distance / Speed of Light

Dominates:
Long distance

Mitigation:
Shorter physical path

Transmission

Packet Size / Link Rate

Dominates:
Slow links

Mitigation:
Higher bandwidth

Processing

Router hardware cycles

Dominates:
Many hops

Mitigation:
Hardware ASICs

Queuing

Buffer wait time

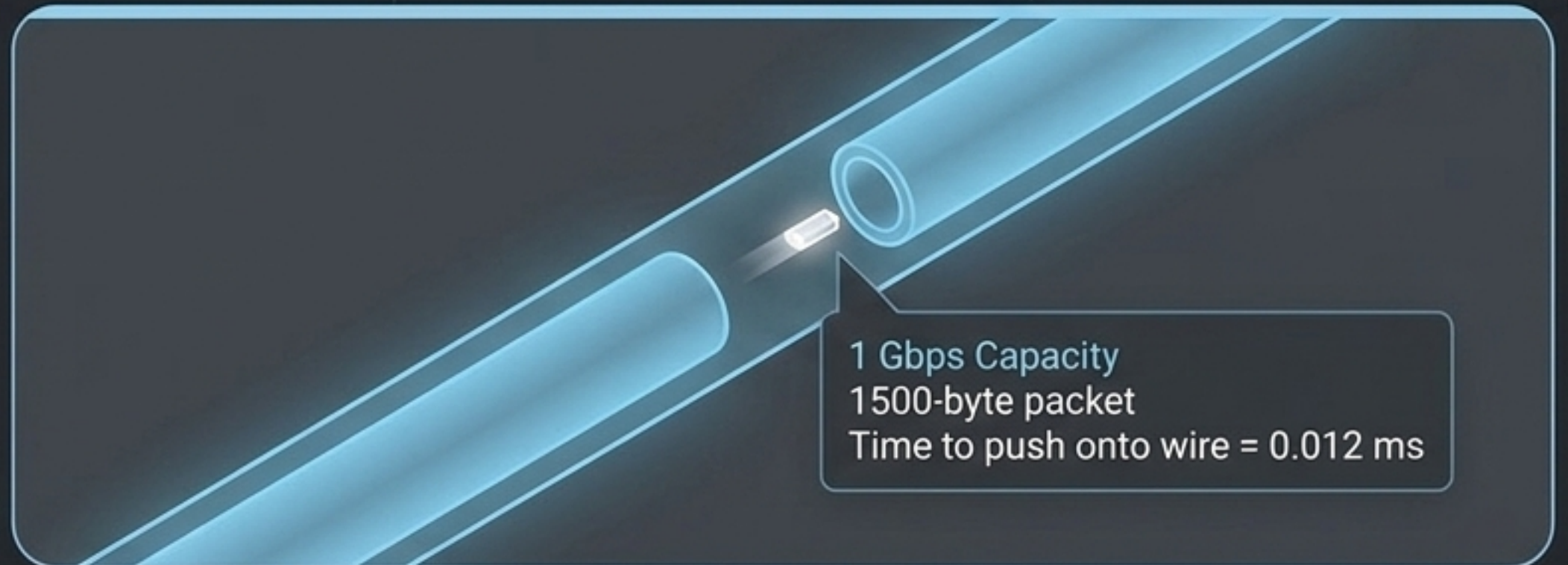
Dominates:
Congested links

Mitigation:
QoS / Traffic reduction

Distance vs. Capacity

On modern gigabit links, Transmission Delay is microscopic.

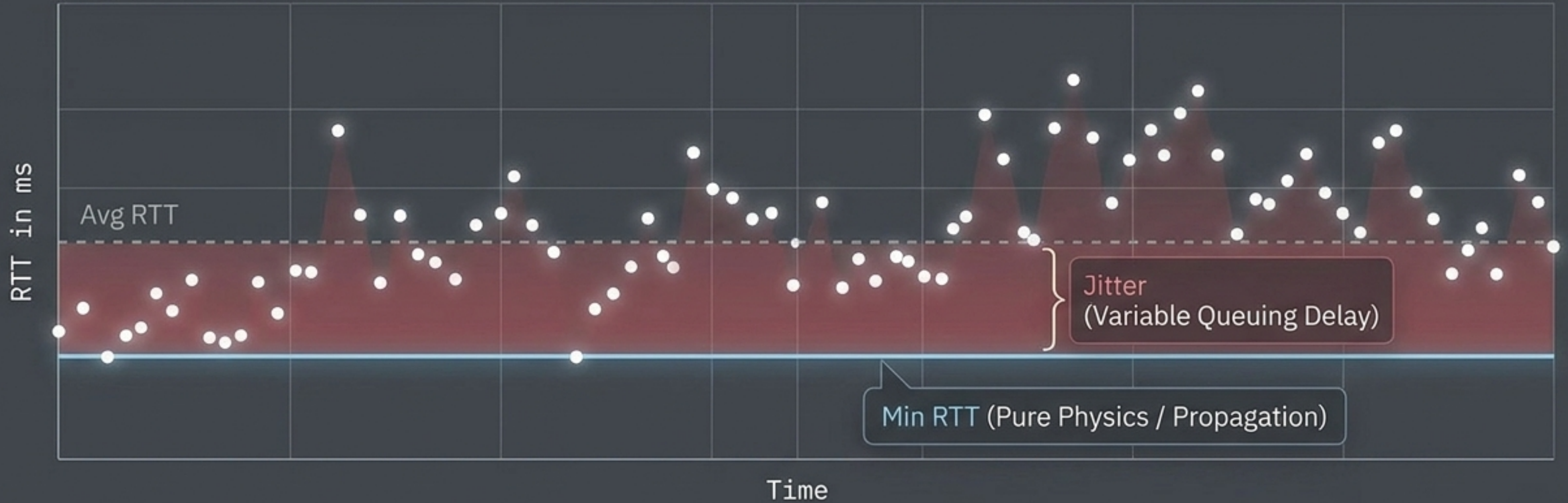
Propagation Delay—the raw speed of light traveling through glass—entirely dominates long-distance communication.



Isolating the Hidden Bottleneck: Queuing

Synthetic scatterplot visualized design.

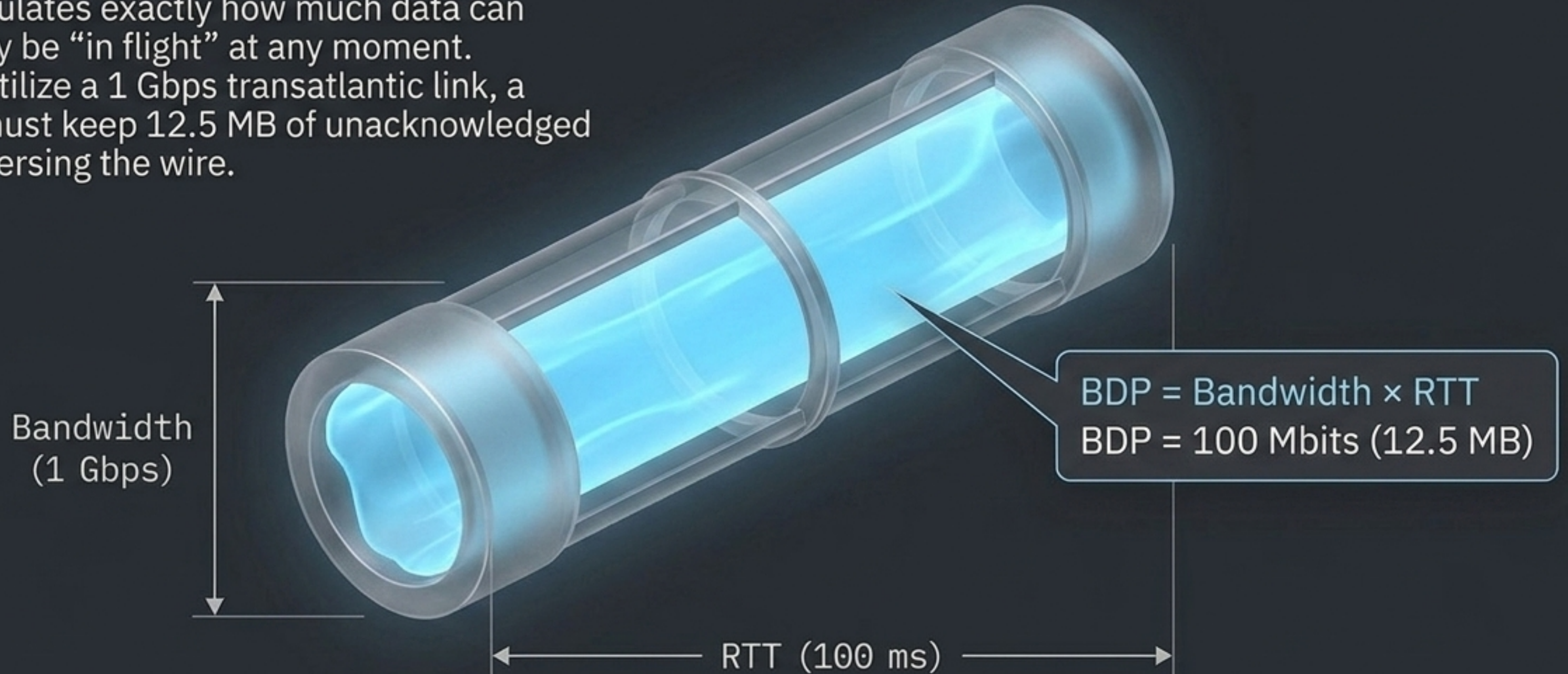
Minimum ping is an unchangeable law of physics. The gap between your minimum and average ping reveals exactly how much time your packets spent waiting in congested router buffers.



The Bandwidth-Delay Product (BDP)

BDP calculates exactly how much data can physically be “in flight” at any moment. To fully utilize a 1 Gbps transatlantic link, a sender must keep 12.5 MB of unacknowledged data traversing the wire.

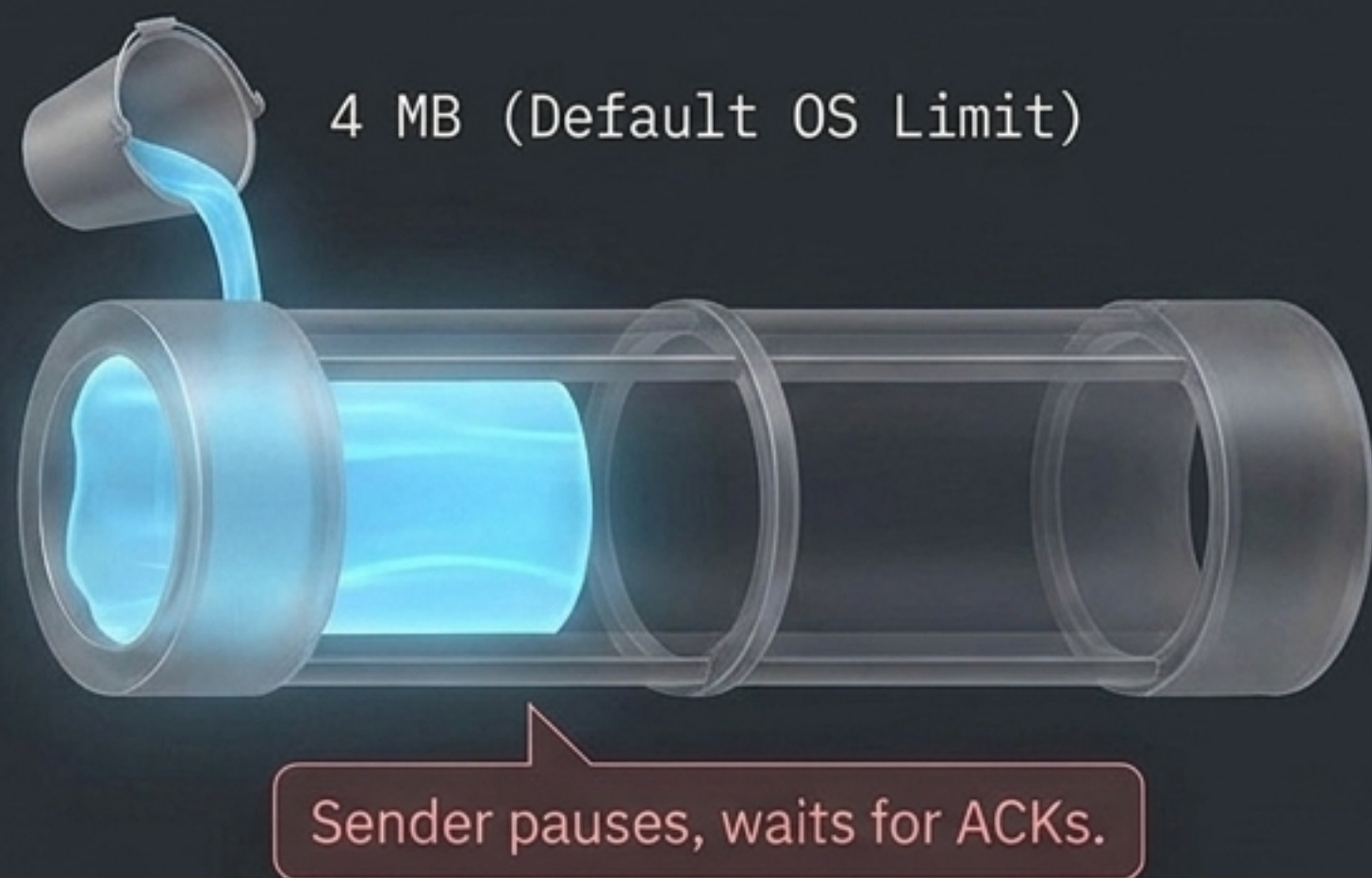
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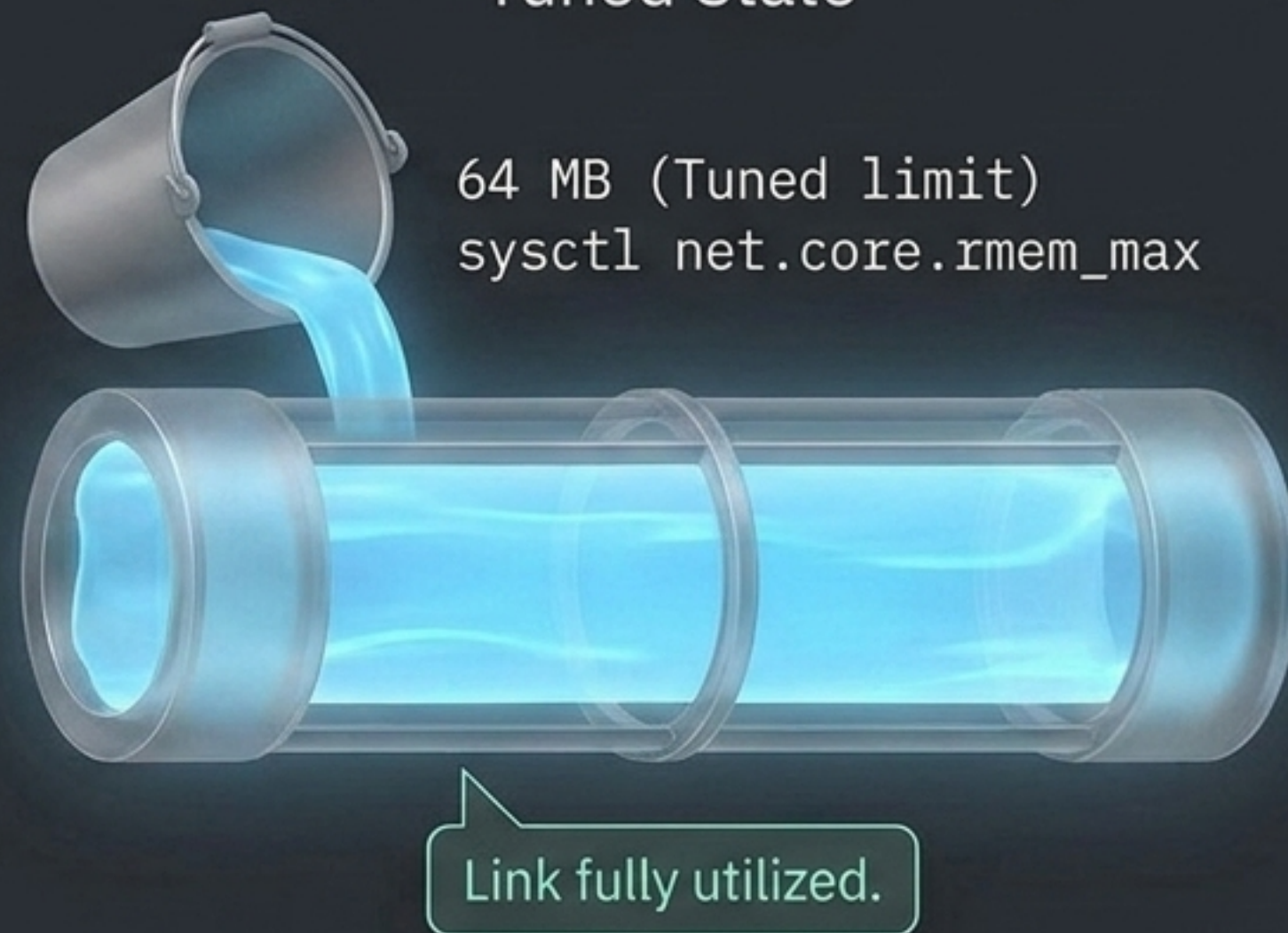
The TCP Buffer Starvation

Throwing more bandwidth at a high-latency link is useless if TCP buffers are too small. The sender will hit its limit, stop transmitting, and wait for acknowledgments—leaving an expensive link sitting idle.

Default State



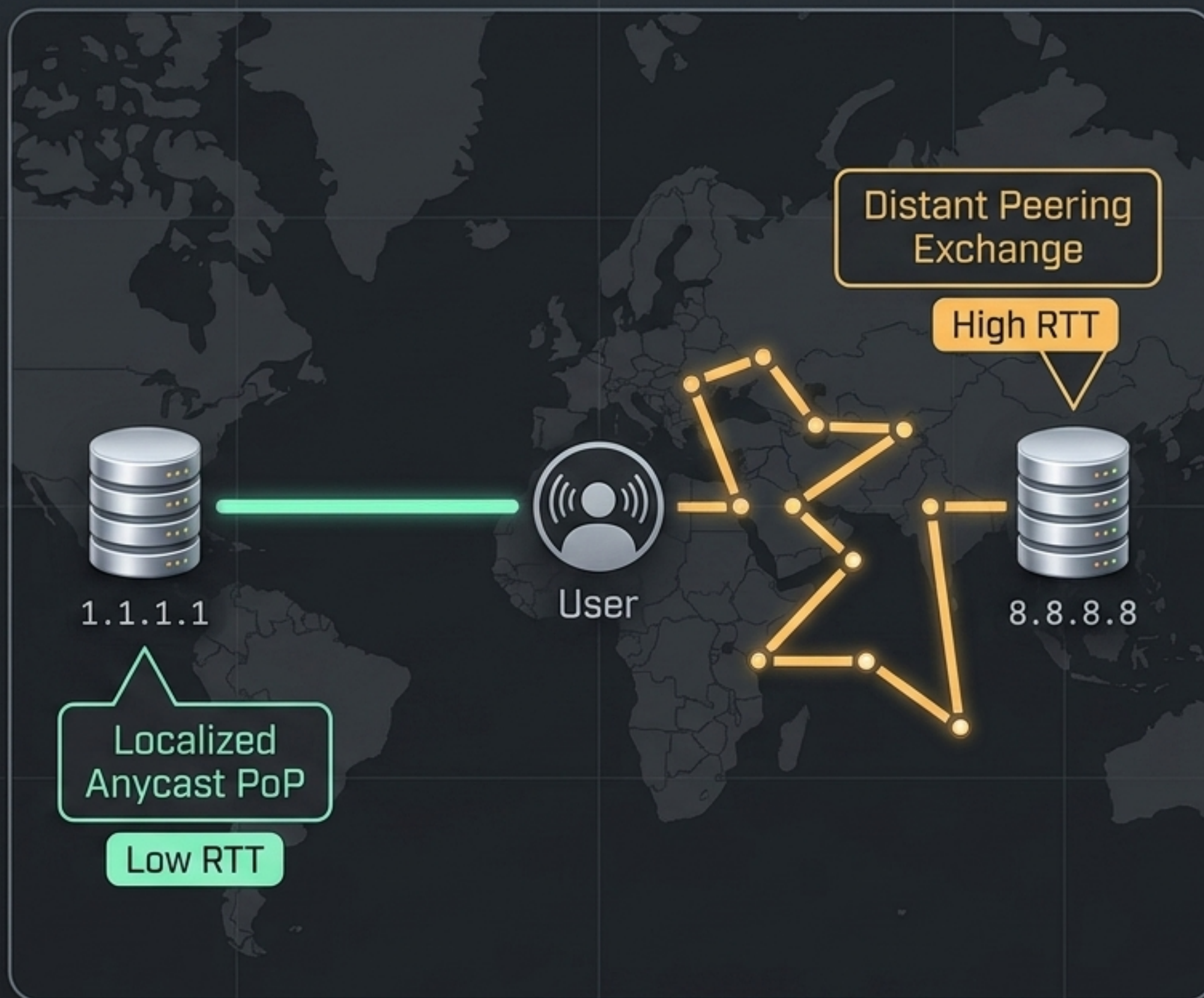
Tuned State



Network Topology ≠ Geographic Distance

Identical IPs and similar geographic distances can yield drastically different ping times.

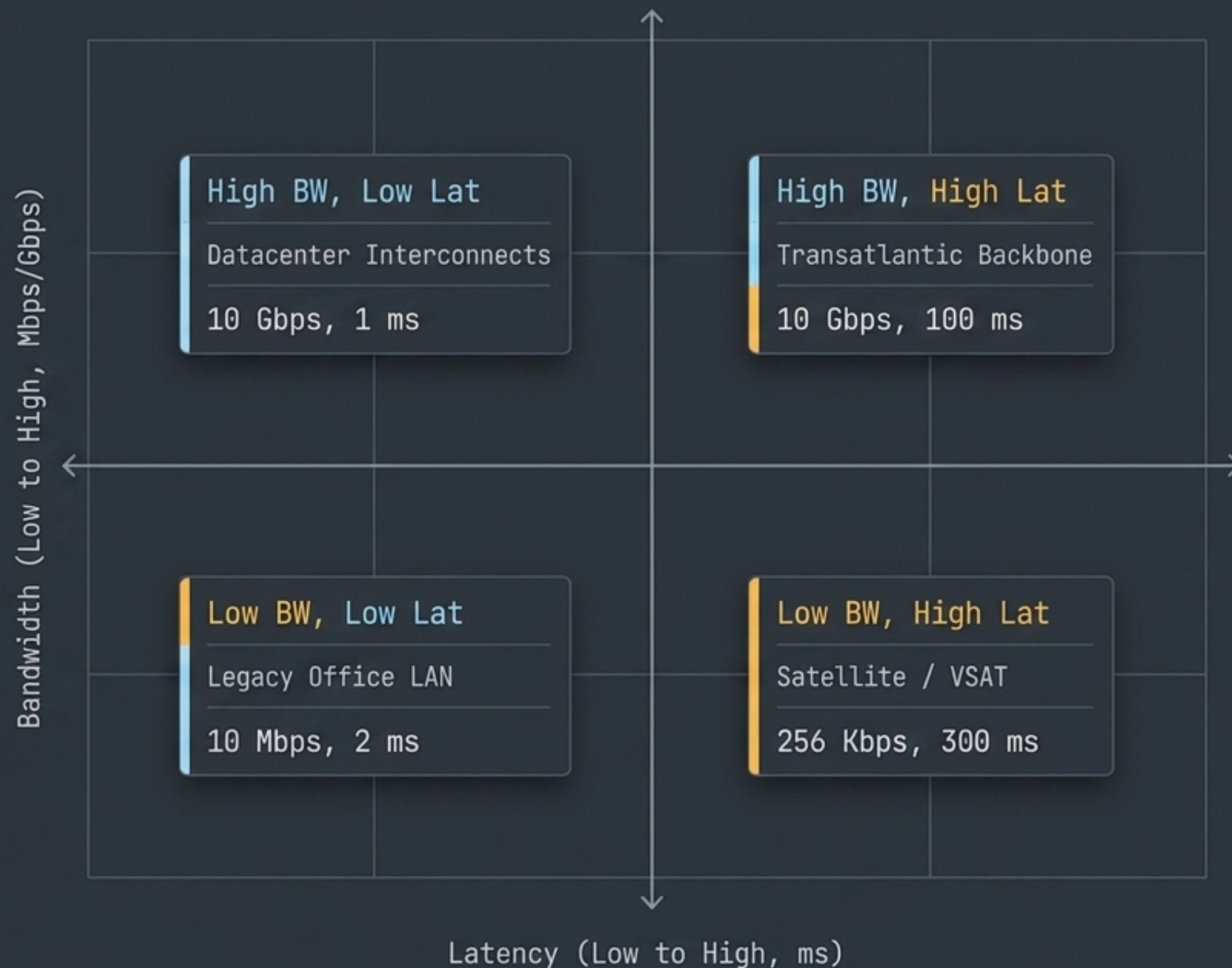
Complex ISP peering agreements and Anycast routing mean internet pathways rarely mirror a physical map.



The Orthogonal Dimensions of Speed

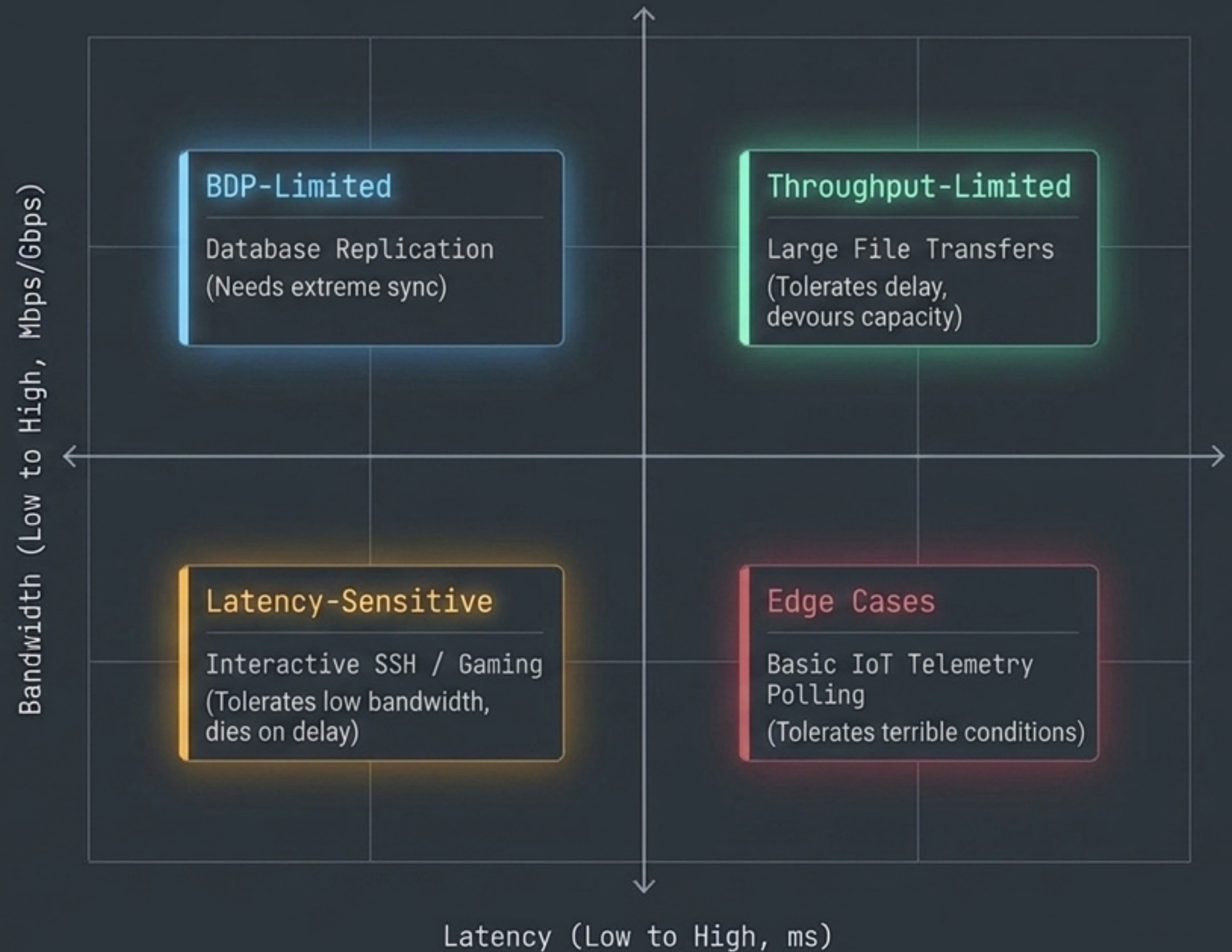
Bandwidth (capacity) and Latency (delay) are entirely independent physical properties.

A loopback interface proves this: infinite bandwidth, near-zero latency, zero physical distance.



Application Realities

Software design is beholden to network physics. Selecting the right protocol requires understanding which quadrant your application is forced to operate within.



The Telemetry Axioms

The defining laws of data transit, distilled from Layer 1 physical constraints up to Layer 4 protocol realities.

1. Signal Strength

SNR alone dictates physical modulation and speed.



2. Delay Profiling

Minimum ping is physics; average ping is traffic.



3. Long Haul Links

Bandwidth scaling requires TCP buffer tuning to match BDP.



4. Routing Reality

Distance on a map rarely matches topology on the wire.



5. The Golden Rule

Bandwidth (capacity) and Latency (delay) are completely orthogonal dimensions.

